Kappa Foundation investment policy

Payment bonds plus USMV

Protect the scheduled payments and pursue the 2030 real-value goal

USD 5.64m payment bonds
USD 6.35m growth allocation
USD 9,169 assumed initial costs

Original case only. The real-value goal and a 15% maximum-loss ceiling are not guaranteed.

The full USD 12 million allocation

Holding	Initial amount	Purpose
Beta	$2,214,672.00	USD 2.4m scholarship in 2024
Delta	$3,429,808.70	USD 3.6m learning centre in 2027
USMV	$6,346,350.71	Long-term purchasing power
Costs	$9,168.59	Assumed initial trading costs
Total	$12,000,000.00	100% of initial resources

VTI benchmarks the growth sleeve. The full policy reference combines VTI growth with the dedicated payment cash flows.

Contractual cash flows cover both payments

Date	Payment funding	Surplus retained
2023	No mandatory payment	$121,739
2024	Beta principal pays $2.4m	$121,739
2025 and 2026	No mandatory payment	$243,478 combined
2027	Delta final payment pays $3.6m	$0
2030	Earlier coupons remain in cash	$486,957 total

No equity sale or positive reinvestment rate is needed to meet the original obligations.

Amounts rounded. Cash custody and payment settlement still require operational controls.

Interest rate shocks change value, not the promised receipts

Bond	YTM	Modified duration	+100bp price change
Alpha	4.30%	0.96	-1.00%
Beta	4.10%	1.92	-1.98%
Gamma	3.91%	2.78	-2.84%
Delta	3.81%	4.50	-4.56%
Epsilon	3.90%	6.66	-6.66%

The payment reserve loses about $200,255 after a +100bp curve shift. Its contractual payment coverage stays intact.

Parallel and curve-shape stresses are scenarios, not probability forecasts. Exact repricing checks duration and convexity.

A 15% loss ceiling cannot be promised

Equity shock	Rate shock	Loss on initial wealth
-20%	+100bp parallel	12.25%
-30%	+100bp parallel	17.53%
-40%	+100bp parallel	22.82%

A $1.014m switch from equities to cash meets the specific -30% / +100bp stress. It raises the equity growth hurdle to 5.60%.

That change does not guarantee a drawdown limit. Equity becomes a larger share of remaining assets after payment bonds mature.

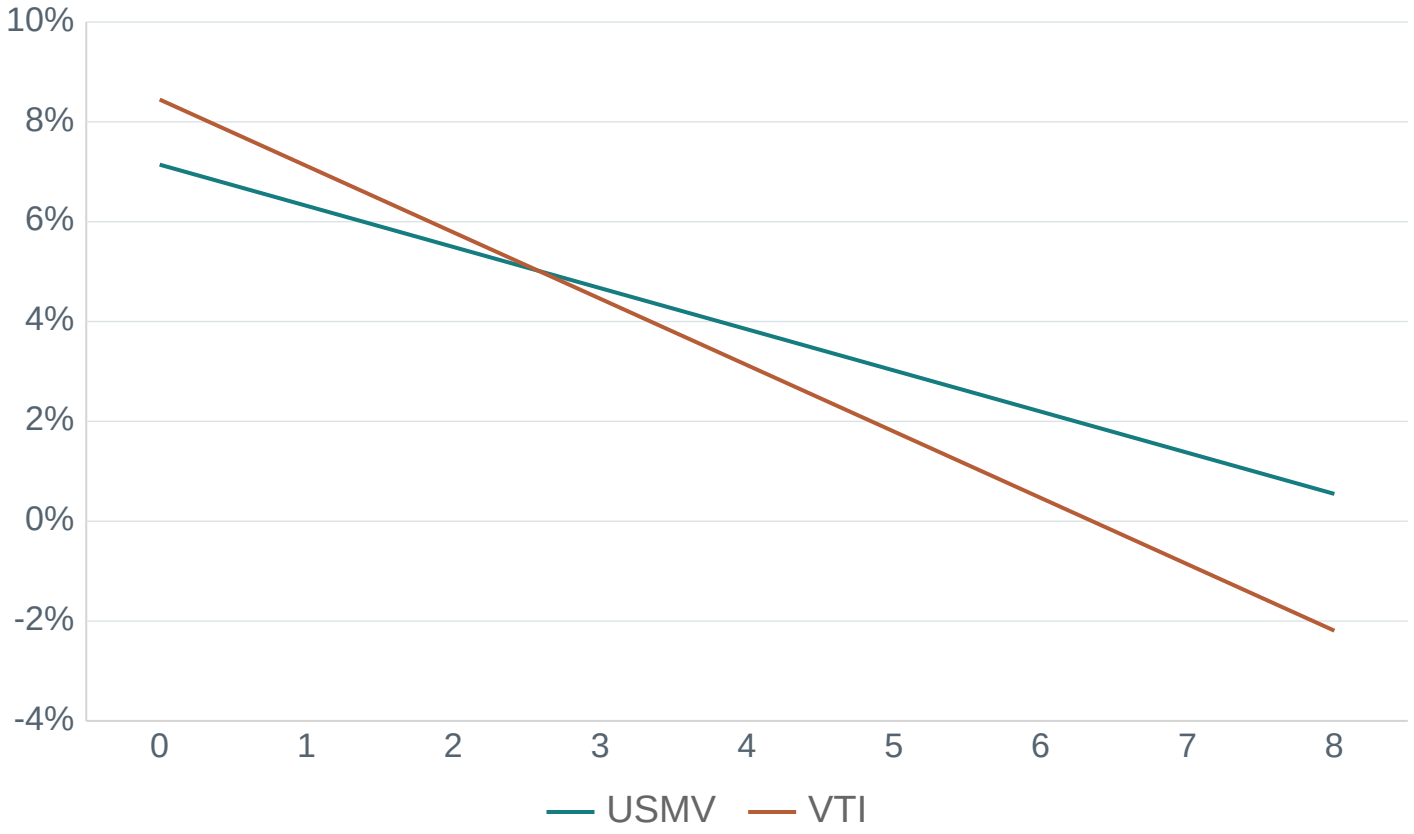
The data separate the decision from its later evaluation

Design choice	Implementation
Candidate universe	8 US stocks, 7 sectors, VTI and USMV
Estimation	96 common monthly returns, Jan 2015 to Dec 2022
Evaluation	36 months, Jan 2023 to Dec 2025
Market source	Python yfinance, saved untransformed daily responses
Factor source	Kenneth French FF3 monthly ZIP, original bytes retained

No invented missing returns. No fictional bond price histories. Current vendor vintages and survivor selection remain limitations.

Dated SEC eligibility evidence is documented. Direct SEC downloads returned 403, so source notes distinguish web verification from saved files.

USMV is preferred at the assumed risk aversion



A = 4
USMV CE 3.84%
VTI CE 3.13%

The ranking reverses below A = 2.58

Horizontal axis: risk-aversion A. Vertical axis: annual certainty equivalent. The preference parameter requires trustee agreement.

Covariance assumptions change the optimized holdings

A = 4 portfolio	VTI	USMV	Direct stocks	Sample volatility
Sample covariance	18.10%	51.90%	30.00%	12.75%
Single-index covariance	11.34%	58.66%	30.00%	12.32%

Changing the risk model implies 14.09% one-way turnover. VTI and USMV residual correlation is 0.63, which the single-index model suppresses.

The sample optimizer adds only about 0.12 percentage points of assumed utility over pure USMV.

The preferred policy avoids direct-stock complexity and concentration. Model fit alone does not justify implementation.

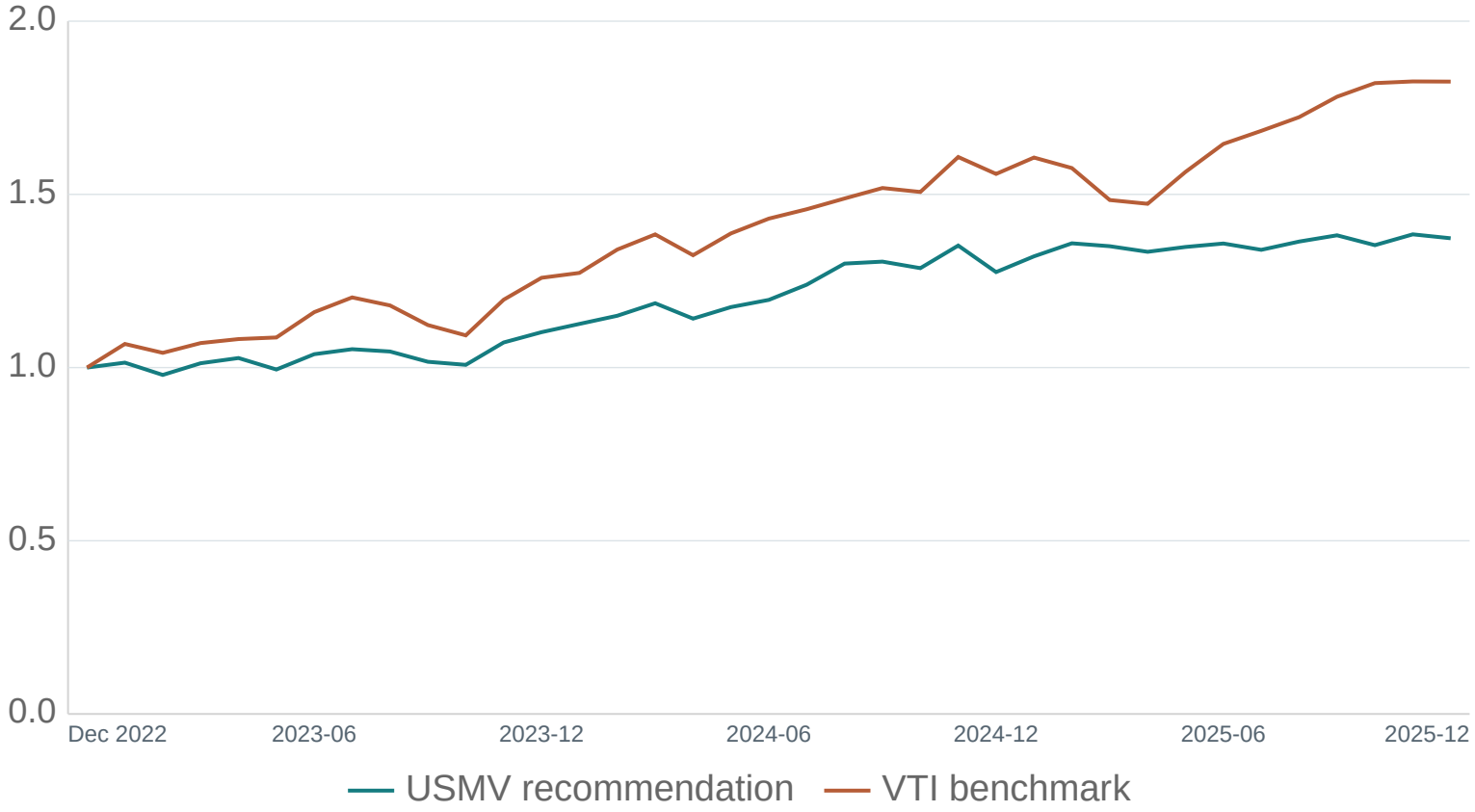
Factor exposure and uncertainty in alpha

Security	Model	Market beta	Annual alpha	95% alpha interval
USMV	CAPM	0.70	2.17%	-1.81% to 6.15%
USMV	FF3	0.74	1.49%	-2.09% to 5.06%
AAPL	CAPM	1.21	12.13%	-1.57% to 25.84%
AAPL	FF3	1.27	11.04%	-1.53% to 23.62%
MSFT	CAPM	0.96	14.47%	4.36% to 24.59%
MSFT	FF3	1.06	12.81%	4.54% to 21.08%

MSFT has positive full-sample alpha. Its early-half interval includes zero, and model and selection uncertainty remain. Forecast alpha stays at zero.

USMV FF3 beta 0.74 and SMB -0.30 indicate lower market sensitivity and a large-company tilt. Its alpha interval includes zero.

USMV substantially lagged VTI in the frozen evaluation



2023 to 2025

USMV +37.33%
VTI +82.56%

USMV also had the lower evaluation Sharpe ratio

Growth of USD 1, net of assumed trading costs. Lower volatility and month-end drawdown came with substantial opportunity cost.

The 2030 real-value goal remains uncertain

USMV scenario	Base	Higher inflation
Annual inflation	2.5%	4.0%
Required equity CAGR	4.84%	6.45%
Modeled chance of real goal	68.5%	48.1%
Median real 2030 assets	$8.89m	$7.91m

Even in the base scenario, about 31.5% of paths miss the goal. Parameter uncertainty is larger than Monte Carlo sampling noise.

Forecast equity arithmetic return 7.14%, annual volatility 12.84%. Historical mean returns are not used as forecasts.

The all-bond alternative makes the compromise explicit

Choice	Outcome under 2.5% inflation
USD 12m in dedicated case bonds	$9.102m fixed nominal terminal cash
Implied 2030 purchasing power	$7.471m in December 2022 dollars
Gap from the USD 8m real goal	6.62% lower real target
Extra initial capital to fund the full target	$480,443

We recommend payment bonds plus USMV to retain the chance of reaching the full real goal, accepting investment risk and possible shortfall.

Nominal cash flows are fixed under case assumptions. Real value depends on inflation and interim bond prices can still fall.

Monitoring and the revised mandate

Quarterly review

Payment coverage and cash custody
Unitized losses excluding programme payments
ETF mandate, concentration and costs
Education-cost inflation and goal feasibility

22 October: preserve the original decision
23 October: quantify the actual mandate change
29 October: submit the final package

AI assistance is disclosed. The group must confirm course policy, add real contributions, and independently check the analysis.